In a Small Transformer, Warmup Matters but Attention Scaling Does Not

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Abstract

The original Transformer relies on two optimization details that are each described in a single sentence: a learning-rate warmup and the scaling of attention logits by $1/\sqrt{d_k}$. Both are copied into most implementations, yet it is rarely tested which of them a given model needs. We ablate both factors in a controlled 2×2 design on a small post-LayerNorm encoder–decoder Transformer trained to reverse digit sequences, with 3 seeds per configuration. Without warmup, every run fails to learn: exact-match accuracy is 0.0% and the training loss stays near 2.34. A control that removes only the ramp, starting at the warmup's peak learning rate, solves the task in 1 of 3 seeds, so the failure is not explained by the large initial learning rate alone. Removing the attention scaling has no measurable effect on the outcome: accuracy is 100.0% without it and 99.8% with it. At this scale ($d_k = 16$), warmup matters and attention scaling does not; whether the scaling matters for larger key dimensions remains untested.

1 Introduction

Vaswani et al. [2017] describe several training details in a single sentence each, and two of them exist to keep optimization stable. The first is a learning-rate schedule that increases linearly for a number of warmup steps before decaying with the inverse square root of the step. The second is a factor of $1/\sqrt{d_k}$ on the attention logits, motivated by the observation that dot products grow with the key dimension d_k and push the softmax into regions with small gradients. Both details are now standard. For a practitioner building a small Transformer, however, it is not obvious which of the two actually matters.

We isolate the two with a 2×2 ablation on a task where success is unambiguous: reversing digit sequences with a small encoder–decoder Transformer. The architecture, data, optimizer, seeds and training budget are held fixed, and only warmup and attention scaling change (Figure 1); a follow-up control separates the warmup ramp from the size of the initial learning rate. Our contributions are:

- **Warmup matters in this setting.** All 6 runs without warmup, with or without attention scaling, fail to learn the task, and a control that removes only the ramp while capping the learning rate at the warmup peak solves it in just 1 of 3 seeds (Section 4, Figure 2).
- **Attention scaling does not.** Removing it leaves accuracy, attention alignment and final loss essentially unchanged (Table 1).
- **The successful runs learn the expected mechanism.** With warmup, an encoder–decoder attention head attends to the source token being copied at almost every target position (Section 4).

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	scaled $QK^\top/\sqrt{d_k}$	unscaled $QK^\top$
warmup w = 200	Baseline	No scaling
no warmup w = 1	No warmup	Neither

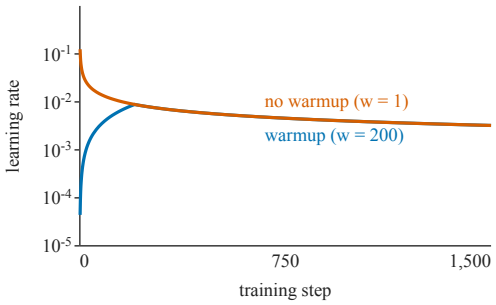


Figure 1: The ablation crosses warmup with attention scaling (left). Without warmup, the learning-rate schedule of Vaswani et al. [2017] starts at its maximum instead of ramping up (right; $d_{\text{model}} = 64$, log scale).

2 Related work

The warmup schedule and the $1/\sqrt{d_k}$ scaling were introduced together with the Transformer [Vaswani et al., 2017]. Popel and Bojar [2018] collect practical training tips for Transformer translation models, including the choice of learning rate and warmup. Xiong et al. [2020] study where layer normalization is placed and show that the pre-LayerNorm variant can be trained without warmup, whereas the original post-LayerNorm design relies on it. Liu et al. [2020] relate the need for warmup to the high variance of the adaptive learning rate of Adam [Kingma and Ba, 2015] early in training. We keep the original post-LayerNorm design and ask a complementary question: how warmup compares with attention scaling when both are ablated together.

3 Experimental setup

Task. Each input is a sequence of 8 tokens drawn uniformly from ten digit symbols, and the target is the same sequence reversed. The vocabulary has 12 tokens (the digits plus padding and a start token). Training batches of 64 sequences are sampled fresh at every step; evaluation uses 500 newly sampled sequences.

Model. An encoder–decoder Transformer following Vaswani et al. [2017]: post-LayerNorm residual blocks, sinusoidal positional encodings, tied input and output embeddings, and Xavier-uniform initialization. The encoder and decoder each have two layers with $d_{\text{model}} = 64$, $h = 4$ attention heads ($d_k = 16$) and feed-forward width 128, without dropout.

Training. We use Adam [Kingma and Ba, 2015] with $\beta_1 = 0.9$, $\beta_2 = 0.98$ and $\epsilon = 10^{-9}$, the learning-rate schedule $\eta_t = d_{\text{model}}^{-1/2} \min\big(t^{-1/2},\, t\, w^{-3/2}\big)$ of Vaswani et al. [2017] with w warmup steps, label smoothing of 0.1 [Szegedy et al., 2016], and 1,500 training steps.

Ablation. We cross two factors. *Warmup*: $w = 200$, or $w = 1$, which removes the ramp so the learning rate starts at 0.125 instead of peaking at 8.8×10^{-3} after 200 steps. *Attention scaling*: logits $QK^\top/\sqrt{d_k}$, or unscaled $QK^\top$. Each of the four configurations is trained with 3 seeds (12 runs); a given seed sees the same initialization and data in every configuration. The *follow-up control* (*no ramp, capped*) starts at the warmup peak 8.8×10^{-3} and then follows the baseline schedule, $\eta_t = d_{\text{model}}^{-1/2} \min\big(t^{-1/2},\, w^{-1/2}\big)$ with $w = 200$, so it differs from the baseline only during the first 200 steps; it uses scaled attention and the same 3 seeds.

Metrics. We report exact-match accuracy of greedy decoding on the evaluation sequences; *alignment*, the fraction of target positions whose strongest encoder–decoder attention (best layer and head) falls on the source token being copied; and the training loss averaged over the last four logged steps. Tables give the mean over seeds and the range from the lowest to the highest seed.

Table 1: Warmup matters and attention scaling does not. Mean over 3 seeds, with the range from the lowest to the highest seed in brackets; the best mean in each column is in bold. *Capped*: no ramp, learning rate capped at the warmup peak (follow-up control).

Configuration	Warmup	Scaling	Accuracy (%)	Alignment (%)	Final loss
Baseline	yes	yes	99.8 [99.4, 100.0]	**100.0** [100.0, 100.0]	**0.53** [0.53, 0.53]
No scaling	yes	no	**100.0** [100.0, 100.0]	**100.0** [100.0, 100.0]	**0.53** [0.53, 0.53]
No warmup	no	yes	0.0 [0.0, 0.0]	13.2 [12.7, 13.6]	2.33 [2.27, 2.36]
Neither	no	no	0.0 [0.0, 0.0]	12.8 [12.5, 13.2]	2.28 [2.26, 2.31]
No ramp, capped	capped	yes	32.9 [0.0, 98.6]	44.1 [17.1, 95.3]	1.51 [0.55, 2.07]

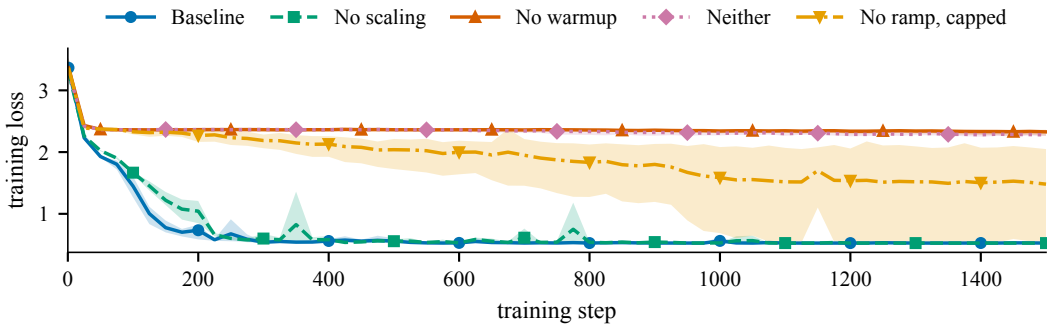


Figure 2: Without warmup, training stalls almost immediately. Training loss for the four configurations and the follow-up control (mean over 3 seeds; shaded: range over seeds). With warmup, the two mean curves nearly coincide after early training; the shaded bands show brief loss spikes in single seeds. The capped no-ramp control improves slowly and solves the task in 1 of 3 seeds.

Compute. All runs used a laptop CPU with 8 PyTorch threads (PyTorch 2.2.2). A single run took between 51 and 66 seconds, about 12 minutes for all runs. The follow-up control ran on the same machine while other jobs were running, and its runs took between 47 and 1051 seconds.

4 Results

Warmup matters. The four configurations of Table 1 split cleanly by warmup. With warmup, every run solves the task: the baseline reaches 99.8% exact-match accuracy (range 99.4–100.0%). Without warmup, all 6 runs fail, with accuracy of 0.0% with scaling and 0.0% without it. Their loss stops improving by step 25 and stays between 2.25 and 2.46 for the rest of training, while the runs with warmup reach a final loss of 0.53.

The ramp itself matters, not only the initial learning rate. Without warmup the learning rate starts at 0.125, far above the warmup peak, so the failures above could come from the large learning rate rather than from the missing ramp. The follow-up control removes the ramp but never exceeds the warmup peak. It solves the task (accuracy of at least 90%) in 1 of 3 seeds, with mean accuracy 32.9% (range 0.0–98.6%); in the other 2 seeds the loss falls only slowly and ends far above that of the baseline (Figure 2). The ramp therefore matters at this learning rate, but it is not strictly necessary for every seed. A rerun of the baseline with the control's code reproduced the original baseline (largest difference in accuracy or final loss: 0.000).

Attention scaling does not change the outcome here. With warmup, removing the scaling gives 100.0% accuracy and a final loss of 0.53 (baseline: 0.53), and the loss curves differ only early in training and in brief loss spikes of single seeds (Figure 2). Without warmup, keeping the scaling does not rescue training either.

The successful runs learn the expected alignment. In both configurations with warmup, an encoder–decoder attention head attends to the source token being copied at 100.0% (baseline) and

100.0% (no scaling) of target positions. For the failed runs it is 13.2%, close to the 12.5% expected if the strongest attention fell on a random source position.

5 Limitations

This is a small, synthetic study: one task, one model size, 1,500 training steps and 3 seeds per configuration. We did not vary the key dimension, so our results say nothing about whether attention scaling becomes necessary for larger d_k, which is where the motivation of Vaswani et al. [2017] applies most strongly. We tested only the post-LayerNorm architecture and a single warmup length, and we did not search over learning rates: pre-LayerNorm models are known to train without warmup [Xiong et al., 2020], and our control tested a single learning-rate cap (the warmup peak) with 3 seeds and the same training budget. Its mixed outcome suggests sensitivity to the seed, and a lower cap or longer training might let more runs without a ramp succeed.

6 Conclusion

In a controlled 2×2 ablation on a small post-LayerNorm Transformer, no run learned sequence reversal without learning-rate warmup, and removing only the ramp left 2 of 3 seeds unsolved, while removing the $1/\sqrt{d_k}$ attention scaling had no measurable effect on the outcome. In settings like ours, dropping warmup is the change to avoid; testing whether attention scaling matters at larger key dimensions is the natural next step.

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